

Lab 172: [Desafío] Ejercicio de instancias de Amazon EC2

- **Dificultad:** ☆☆☆ (Intermedia / Desafío)
- **Tiempo Estimado:** ⌚ 45 minutos
- **Servicios Principales:** 🌐 Amazon VPC, 🖥️ Amazon EC2, 🛡️ Security Groups, 🌐 Internet Gateway (IGW).

1. Resumen y Objetivos

Este laboratorio de desafío requiere la implementación de una pila tecnológica completa para una aplicación web estática. A diferencia de los laboratorios guiados, deberás construir los cimientos de red (VPC) y conectividad antes de desplegar el poder de cómputo.

Objetivos del desafío:

- 🏗️ **Diseñar** una red virtual aislada y personalizada.
- 🌐 **Habilitar** conectividad bidireccional con el Internet público.
- 🖥️ **Aprovisionar** una instancia de cómputo Amazon Linux optimizada.
- 📜 **Automatizar** la configuración del servidor web mediante scripts de arranque.
- 🚀 **Desplegar** contenido personalizado y verificar la alta disponibilidad.

2. Análisis del Escenario

El diagnóstico técnico indica que la aplicación requiere un entorno "VPC-Only" para garantizar el aislamiento. El éxito del despliegue depende de tres factores críticos: la correcta asociación del **Internet Gateway** a la tabla de rutas, la asignación de una **IP pública dinámica** y la ejecución impecable del script de **User Data** para evitar configuraciones manuales post-lanzamiento.

3. Arquitectura

- **VPC:** 10.0.0.0/16.
 - **Subred Pública:** 10.0.1.0/24.
 - **Enrutamiento:** Tabla de rutas con destino 0.0.0.0/0 hacia el IGW.
 - **Instancia:** Amazon Linux (t3.micro) con almacenamiento General Purpose SSD (gp2).
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4. Desarrollo de las Tareas Paso a Paso

🕒 Fase de Red: Creación de la VPC y Conectividad

1. **Navega** al servicio **VPC** desde la consola de AWS.
2. **Haz clic** en **Create VPC**. Selecciona la opción **VPC only**.
 - **Name tag:** Challenge-VPC.
 - **IPv4 CIDR block:** 10.0.0.0/16.
 - **Haz clic** en **Create VPC**.

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional Info
Create a tag with a key of 'Name' and a value that you specify.

Challenge-VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info
Default

VPC encryption control (S) Info
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode
See which resources in your VPC are unencrypted but allow the creation of unencrypted resources. ☐ Enforce mode
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q Challenge-VPC X Remove tag

Add tag
You can add 49 more tags

Cancel Preview code Create VPC

3. Crea la Subred: En el panel izquierdo, selecciona **Subnets** -> **Create subnet**.

- **VPC ID:** Selecciona **Challenge-VPC**.
- **Subnet name:** **Public-Subnet-Challenge**.
- **IPv4 CIDR block:** **10.0.1.0/24**.
- **Haz clic en Create subnet.**

Create subnet Info

VPC

VPC ID Info
Create subnets in this VPC.

vpc-031fda53b0ec4a62 (Challenge-VPC)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings Info
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name Info
Create a tag with a key of 'Name' and a value that you specify.

Public-Subnet-Challenge
The name can be up to 255 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24
256 IP

Tags - optional

Key
Q Name X

Value - optional
Q Public-Subnet-Challenge X Remove

Add new tag
You can add 49 more tags

Remove

Add new subnet

4. Crea el Internet Gateway: Selecciona **Internet gateways** -> **Create internet gateway**.

- **Name tag:** **Challenge-IGW**.
- **Haz clic en Create internet gateway.**
- **Selecciona** el IGW recién creado, haz clic en **Actions** -> **Attach to VPC** y elige tu **Challenge-VPC**.

Create internet gateway [info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Challenge-IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	Challenge-IGW	Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

Attach to VPC (igw-091c093808c8438d5) [info](#)

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

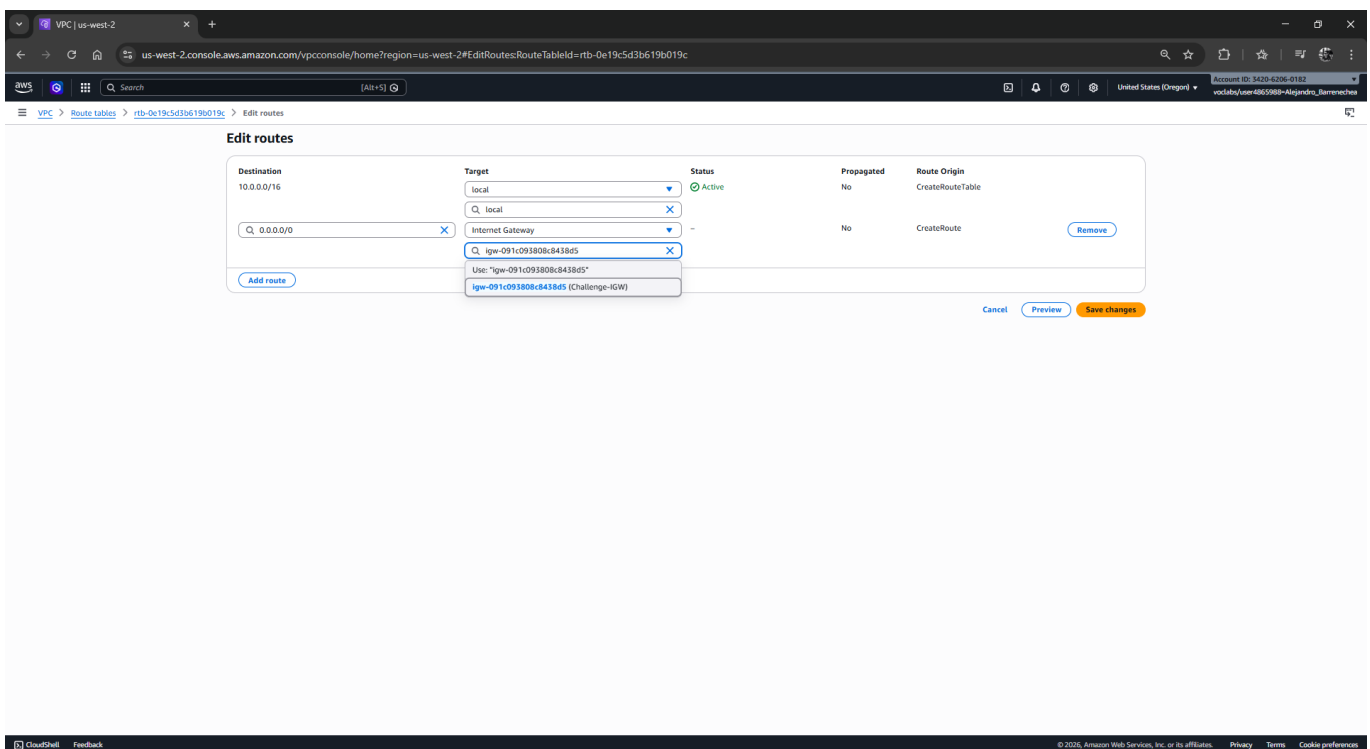
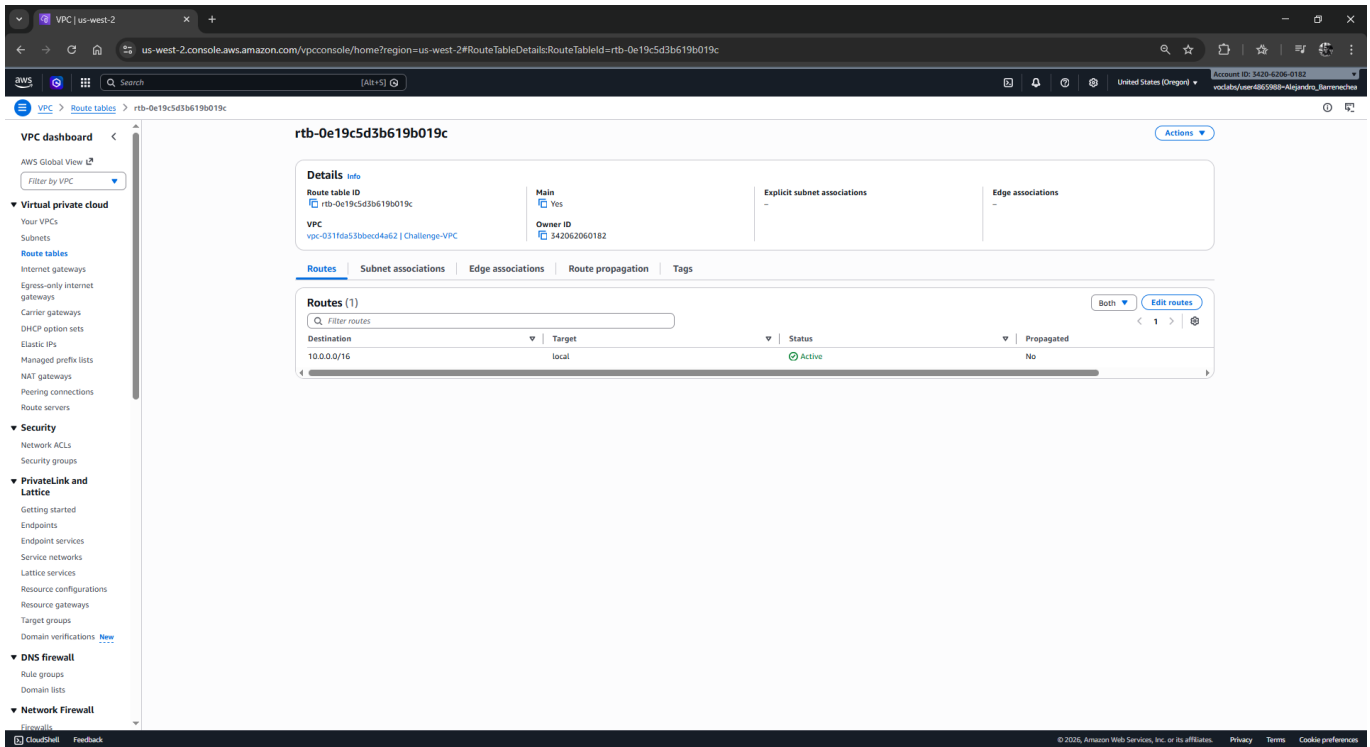
Available VPCs
Attach the internet gateway to this VPC.

vpc-031fda53b0ec4a62	
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[AWS Command Line Interface command](#)

[Cancel](#) [Attach internet gateway](#)

5. **Configura** la Tabla de Rutas: Selecciona **Route tables**. Busca la tabla asociada a tu **Challenge-VPC**.
 - **Haz clic** en el ID de la tabla -> pestaña **Routes** -> **Edit routes**.
 - **Añade una ruta:** Destination **0.0.0.0/0** | Target: **Internet Gateway** -> Selecciona **Challenge-IGW**.
 - **Guarda** los cambios.



2 Fase de Cómputo: Lanzamiento de la Instancia EC2

1. **Navega** al servicio **EC2** y haz clic en **Launch instance**.
2. **Nombre y Etiquetas:** Escribe **Web-Server-Challenge**.
3. **Imagen (AMI):** Selecciona **Amazon Linux 2023** (o Amazon Linux 2).
4. **Tipo de Instancia:** Busca y selecciona **t3.micro**.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name:

Application and OS Images (Amazon Machine Image)

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-043ab4148b7bb33e9 (64-bit (x86), self-preferred) / ami-004782703d687d5d1 (64-bit (x86), self-preferred)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260330.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	self-preferred	ami-043ab4148b7bb33e9	2026-03-30	ec2-user

Instance type

t3.micro
Family: t3, 2 vCPUs, 1 GiB Memory, 30 GiB Storage, 100 Gbps network, 100 Gbps EBS, 100 Gbps ENA, 100 Gbps CloudWatch Logs

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-043ab4148b7bb33e9

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro Instance usage (or t3.micro where t3.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

[Cancel](#) [Launch instance](#) [Preview code](#)

5. **Key pair (Login):** Selecciona **Proceed without a key pair (Not recommended).**

6. **Network settings (Configuración de red):** Haz clic en **Edit.**

- **VPC:** Selecciona **Challenge-VPC.**
- **Subnet:** Selecciona **Public-Subnet-Challenge.**
- **Auto-assign public IP:** Cambia a **Enable.**
- **Security group:** Selecciona **Create security group.**
 - **Security group name:** **Web-Challenge-SG.**
 - **Regla SSH:** Permite puerto 22 desde **Anywhere (0.0.0.0/0).**
 - **Regla HTTP:** Haz clic en **Add security group rule**, selecciona **HTTP** (puerto 80) desde **Anywhere (0.0.0.0/0).**

Network settings

VPC - required: vpc-031fda53b0ec4a62 (Challenge-VPC)

Subnet: subnet-020504cc6a5e85f (Public-Subnet-Challenge)

Auto-assign public IP: Enable

Firewall (security groups): Create security group

Security group name: Web-Challenge-SG

Description: launch-wizard-1 created 2026-04-02T18:54:19.451Z

Inbound Security Group Rules

Type	Protocol	Port range	Source type	Source	Description - optional
SSH	TCP	22	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop
HTTP	TCP	80	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-043ab4148b7bb33e9

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro Instance usage (or t3.micro where t3.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

[Cancel](#) [Launch instance](#) [Preview code](#)

7. Almacenamiento: Mantén 8 GiB de tipo **gp2**.

▼ **Configure storage** [Info](#) Advanced

1x GiB Root volume, Not encrypted

[Add new volume](#)

🔄 Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

8. Detalles Avanzados (User Data): Despliega **Advanced details**, ve al final y **pega** el siguiente script:

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
# Ajuste de permisos para ec2-user
chown -R ec2-user:ec2-user /var/www/html
chmod -R 775 /var/www/html
```

9. Haz clic en **Launch instance**.

Launch an instance | EC2 | us-west-2 | us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#launchInstances

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...read more
ami-0c3ab4148b-7b615af

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t3.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Changes may apply depending on your account's free tier status.

[Cancel](#) [Launch instance](#) [Preview code](#)

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
# Ajuste de permisos para ec2-user
chown -R ec2-user:ec2-user /var/www/html
chmod -R 775 /var/www/html
```

☐ User data has already been base64 encoded

③ Fase de Aplicación: Despliegue de Contenido

1. **Espera** a que el estado sea **Running** y los "Status Checks" sean **2/2 passed**.
2. **Selecciona** la instancia y haz clic en el botón superior **Connect**.

The screenshot shows the AWS Management Console for the 'us-west-2' region. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, Placement Groups, Key Pairs, Network Interfaces, Load Balancing, Load Balancers, Target Groups, Trust Stores, Auto Scaling, and Auto Scaling Groups. The main content area displays the details for the EC2 instance 'i-0d52c54455e4ea745' (Web-Server-Challenge). The instance is in the 'Running' state, using the 't3.micro' instance type, and has a public IP address of '35.93.115.44'. The 'Connect' button is visible in the top right corner of the instance details panel.

3. Usa la pestaña **EC2 Instance Connect** y haz clic en **Connect**.

The screenshot shows the 'Connect to instance' page in the AWS Management Console. The page title is 'Connect to instance | EC2 | us-west-2'. The main content area is titled 'Connect' and includes a sub-header 'Connect to an instance using the browser-based client.' Below this, there are four tabs: 'EC2 Instance Connect', 'SSM Session Manager', 'SSH client', and 'EC2 serial console'. The 'EC2 Instance Connect' tab is selected. Under this tab, there are two options for connecting: 'Connect using a Public IP' (selected) and 'Connect using a Private IP'. The 'Public IP' option is further detailed with a 'Public IP address' field showing '35.93.115.44'. There is also a 'Username' field with 'ec2-user' entered. At the bottom right, there are 'Cancel' and 'Connect' buttons.

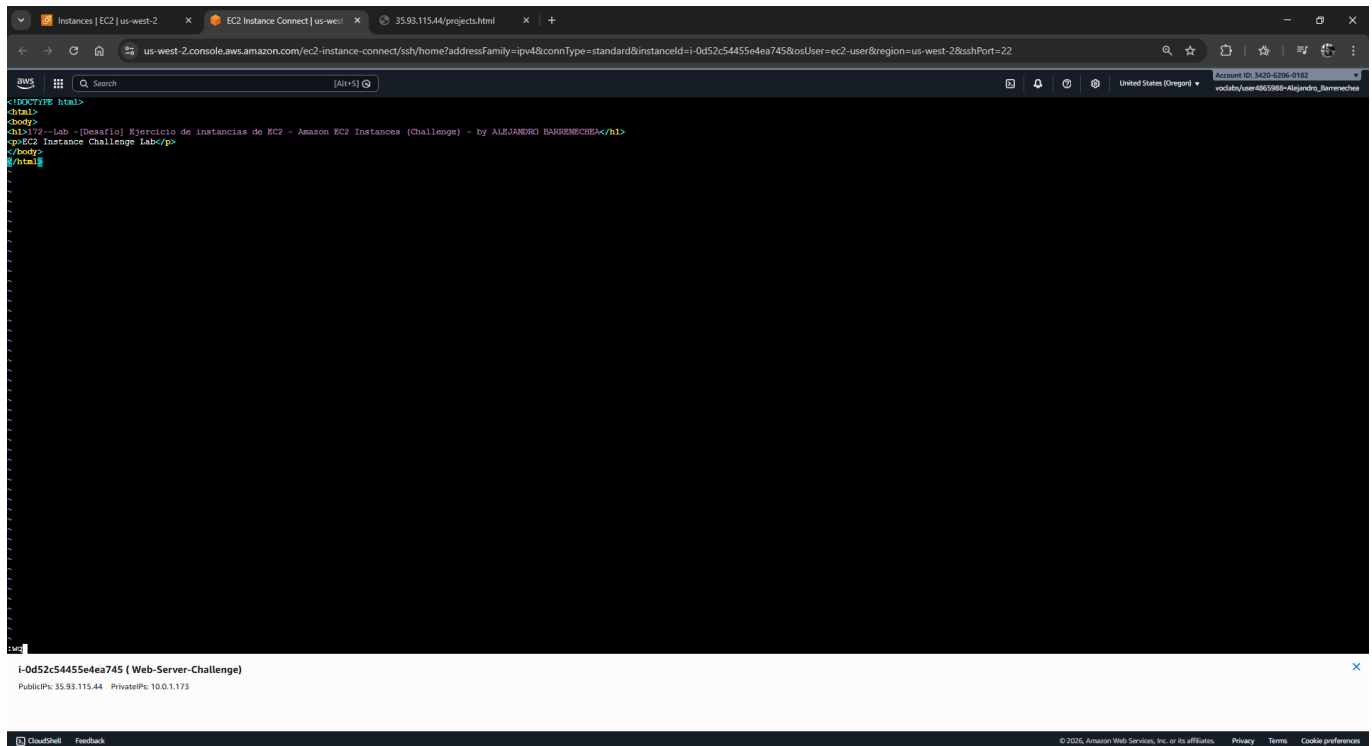
4. Crea el archivo del proyecto mediante el editor **vi**:

```
vi /var/www/html/projects.html
```

5. **Presiona** la tecla **i** e inserta el siguiente código HTML (personaliza con tu nombre):

```
<!DOCTYPE html>
<html>
<body>
<h1>Trabajo del Proyecto re/Start de [TU-NOMBRE]</h1>
<p>EC2 Instance Challenge Lab</p>
</body>
</html>
```

6. Presiona **Esc**, escribe **:wq** y presiona **Enter**.



4 Fase de Verificación y Evidencia

1. **Copia** la **Public IPv4 address** de tu instancia desde la consola EC2.
2. **Abre** una nueva pestaña en tu navegador y navega a: <http://<TU-IP-PUBLICA>/projects.html>.
3. **Captura de Pantalla 1:** Toma una imagen de tu página web funcionando.



EC2 Instance Challenge Lab

-
- The screenshot displays the AWS Management Console's 'Instances' page. At the top, the navigation bar shows the 'Instances' link. The main content area features a table of instances. A single instance, 'Web-Serve-Challenge', is listed with the ID 'i-0d52c54455e4ea745' and is in the 'Running' state. Below the table, the 'Instance details' section provides comprehensive information about the instance, including its public and private IP addresses, instance type ('t3.micro'), and various network and security configurations. The instance is running on the 't3.micro' instance type in the 'us-west-2' region.

Instance diagnostics [Info](#)

This page consolidates information from various AWS sources about your EC2 instance, to help you diagnose and understand instance behavior.

Status overview

Instance state: **Running**

System status check: **Check passed**

Instance status check: **Check passed**

EBS status check: **Check passed**

System log

When you experience issues with your EC2 instance, reviewing system logs can help you pinpoint the cause.

Last updated April 02, 2026, 16:09 (GMT-3) [Copy log](#) [Download](#)

```
[ 33.595865] cloud-init[1634]: verifying : librotll-1.0.9-4.amzn2023.0.2.x86_64 10/13
[ 33.596752] cloud-init[1634]: verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
[ 33.598360] cloud-init[1634]: verifying : mod_httpd-2.0.27-1.amzn2023.0.3.x86_64 12/13
[ 33.598874] cloud-init[1634]: verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13
[ 33.922783] cloud-init[1634]: Installed:
[ 33.923615] cloud-init[1634]: apt-1.7.5-1.amzn2023.0.4.x86_64
[ 33.924693] cloud-init[1634]: apt-util-1.6.3-1.amzn2023.0.2.x86_64
[ 33.925825] cloud-init[1634]: apt-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
[ 33.927001] cloud-init[1634]: apt-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
[ 33.928254] cloud-init[1634]: generic-logos-httpd-16.0.0-1.amzn2023.0.3.noarch
[ 33.929595] cloud-init[1634]: httpd-2.4.66-1.amzn2023.0.3.x86_64
[ 33.930923] cloud-init[1634]: httpd-core-2.4.66-1.amzn2023.0.1.x86_64
[ 33.932175] cloud-init[1634]: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
[ 33.933598] cloud-init[1634]: httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
[ 33.934798] cloud-init[1634]: librotll-1.0.9-4.amzn2023.0.2.x86_64
[ 33.936045] cloud-init[1634]: mailcap-2.1.49-3.amzn2023.0.3.noarch
[ 33.937255] cloud-init[1634]: mod_httpd-2.0.27-1.amzn2023.0.3.x86_64
[ 33.938526] cloud-init[1634]: mod_lua-2.4.66-1.amzn2023.0.1.x86_64
[ 33.939723] cloud-init[1634]: complete!
[ 34.108398] cloud-init[1634]: Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service > /usr/lib/systemd/system/httpd.service.
[ 34.793965] rpm-generator[config@3091]: rpmdb: system has too much memory (910M), limit is 900M, ignoring.
[ 34.794000] rpm-generator[config@3091]: rpmdb: system has too much memory (910M), limit is 900M, ignoring.
```

For boot or networking issues, use the EC2 serial console for troubleshooting. Choose the **Connect** button to start a session.

5. Respuestas Analíticas

1. ¿Por qué se debe habilitar manualmente la "Auto-assign public IP" en este desafío? En una VPC creada manualmente (VPC Only), las subredes por defecto no asignan IPs públicas a las instancias para mantener la seguridad. Como este servidor debe ser accesible desde el navegador, es imperativo forzar esta asignación durante el lanzamiento para obtener una dirección ruteable en internet.

2. ¿Cuál es la importancia del Internet Gateway (IGW) en la tabla de rutas? El IGW actúa como el puente entre la red privada de AWS y la red pública mundial. Sin la entrada **0.0.0.0/0** apuntando al IGW en la tabla de rutas, la instancia puede tener una IP pública, pero los paquetes de datos no sabrán cómo salir de la VPC ni cómo regresar, resultando en un error de "Connection Timeout".

3. ¿Qué beneficio técnico aporta el uso de User Data en comparación con la instalación manual? El User Data garantiza la **idempotencia** y la **automatización**. Permite que la infraestructura esté lista para producir inmediatamente después del arranque, eliminando errores humanos en la consola y permitiendo que, en caso de fallo, se pueda lanzar una nueva instancia idéntica automáticamente (escalabilidad).